Final data analysis

Support, response geometry, language coverage, and item-difficulty diagnostics.

1 Support matters

Sparse item responses create exact 0/1 pass rates that are not reliable difficulty estimates. The first two figures show what the support filter removes.

2 Text and responses are different signals

Language and content length come from item text. Response matrices, empirical pass rates, and classic IRT fits use observed labels and are training-only.

3 Geometry is descriptive

The Qwen kNN graph and its Laplacian describe content structure. They motivate a graph ablation, but do not by themselves prove an online-prediction gain.

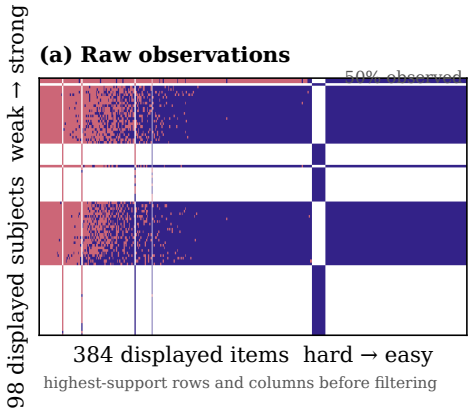
4 Transfer is the open problem

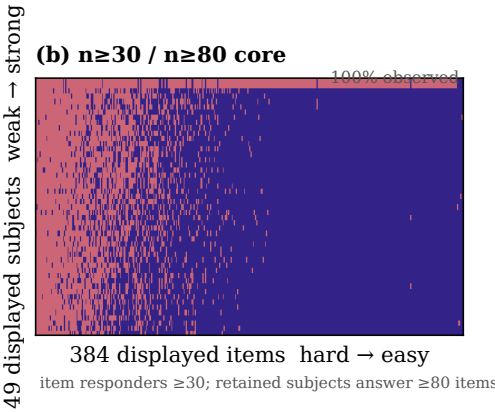
The classic-IRT diagnostic checks recoverability when item responses are available. The archival neural diagnostic tests whether an embedding-only prior recovered comparable item variation.

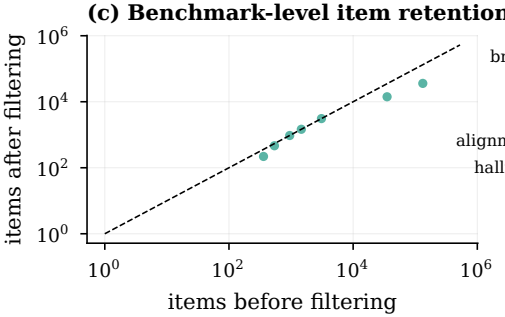
Scope: no leaderboard or acquisition-policy claim here; see the frozen sequential replay.

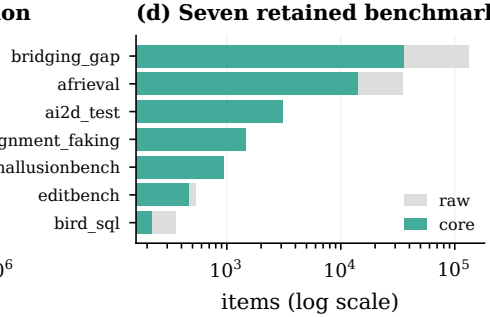
Response matrix before and after support filtering

AfriEval samples: blue = correct, rose = incorrect, white = unobserved.





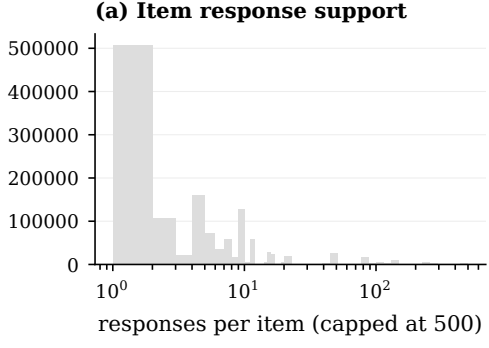


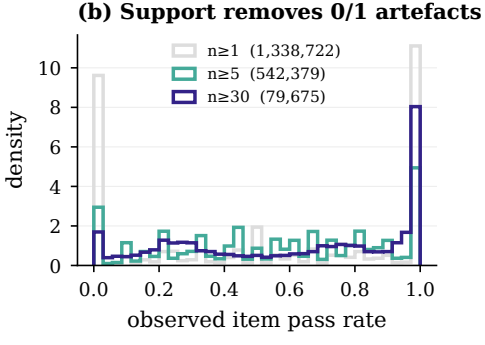


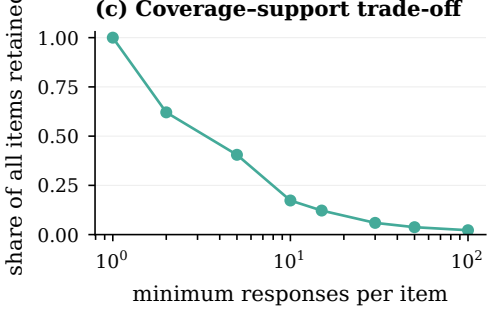
AfriEval: 1,403,380 responses → 670,936; 35,205 items → 14,145; 948 subjects → 898. Corpus core: 54 → 7 benchmarks; 1,338,722 → 56,308 items. Panels are descriptive samples, not a cellwise matched comparison.

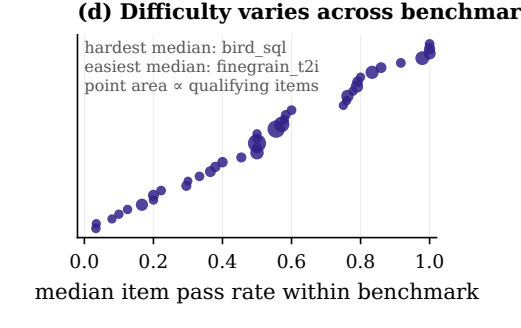
Response support and empirical item difficulty

Pass rate is the observed share correct; one response forces a 0/1 estimate.





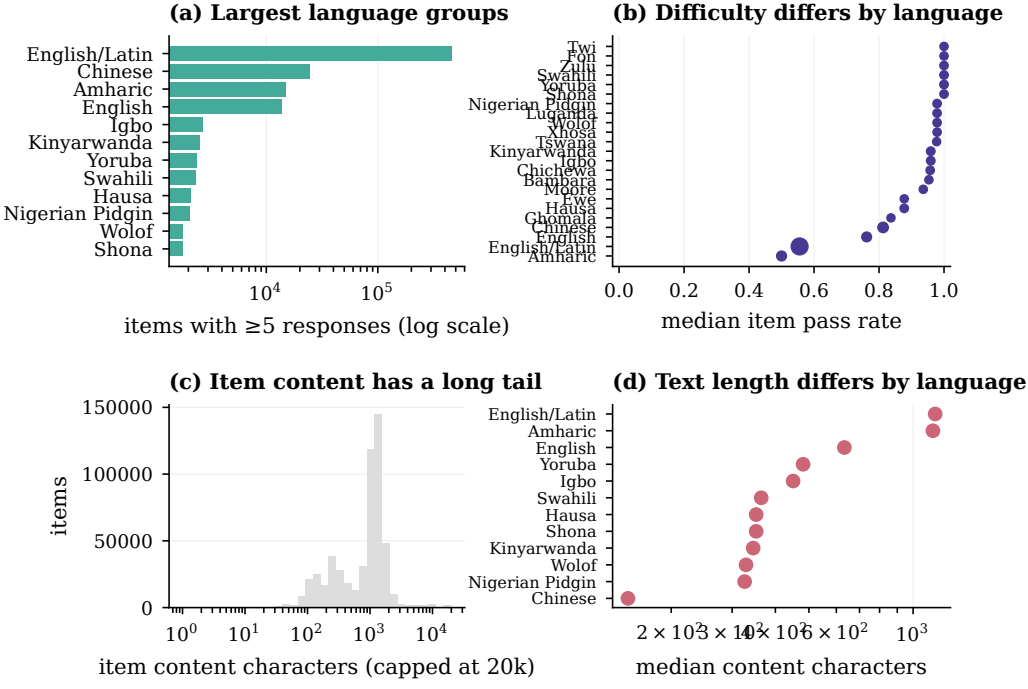




1,338,722 items total; 542,379 have ≥5 responses; 79,675 have ≥30. The support threshold is a measurement-quality decision, not a test-time assumption.

Language and content-length coverage

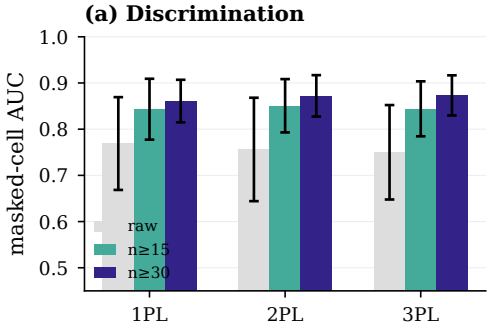
Language is heuristic metadata: a coverage audit, not a causal difficulty label.

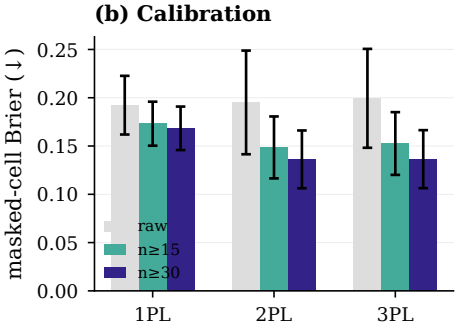


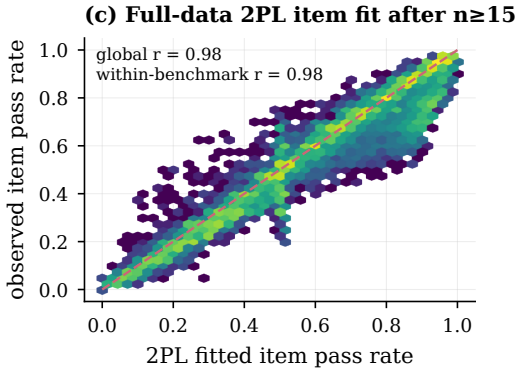
Implication: multilingual, long-context item encoders are sensible inputs. Coverage differences alone do not establish cross-benchmark difficulty transfer.

Filtering and within-benchmark item difficulty

Classic IRT fits observed response matrices and scores masked cells; it cannot cold-start a new item.







What this establishes

- Stable response matrices contain learnable it

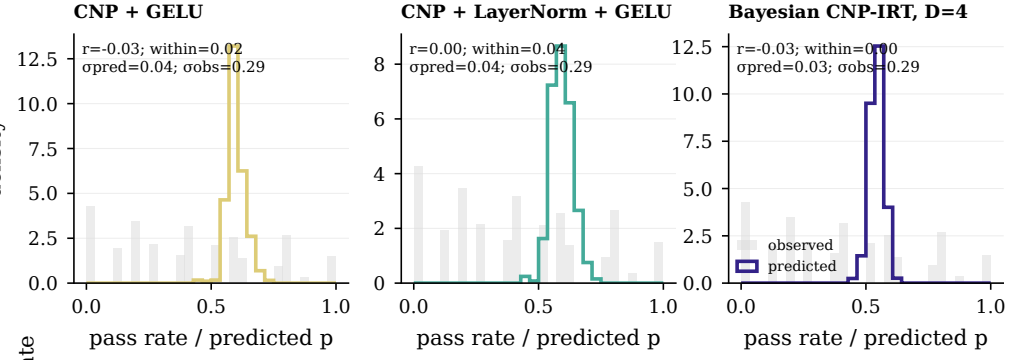
What it does not establish

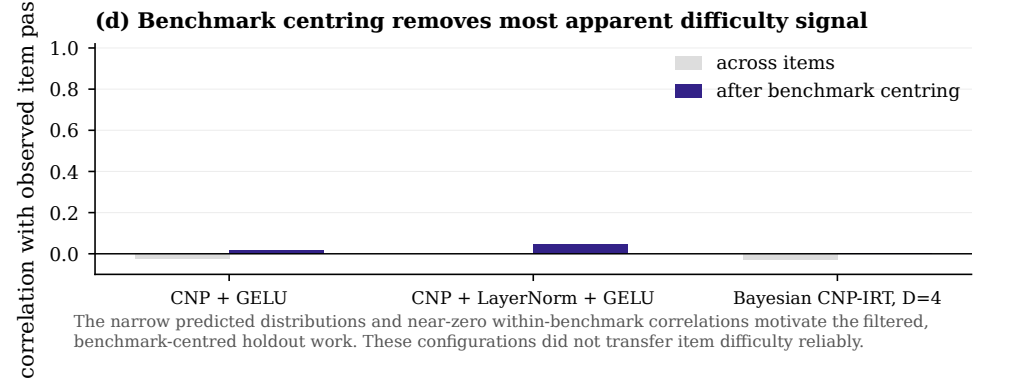
- Text embeddings predict difficulty for a cold
- A test-time submission can use these fitted pa

Coverage: n≥15 retains 17 benchmarks; n≥30
The stricter filter is a quality–coverage trade-o

Archival neural item-prior diagnostic

Historical non-holdout split, test items at budget 0; not the final benchmark-holdout result.

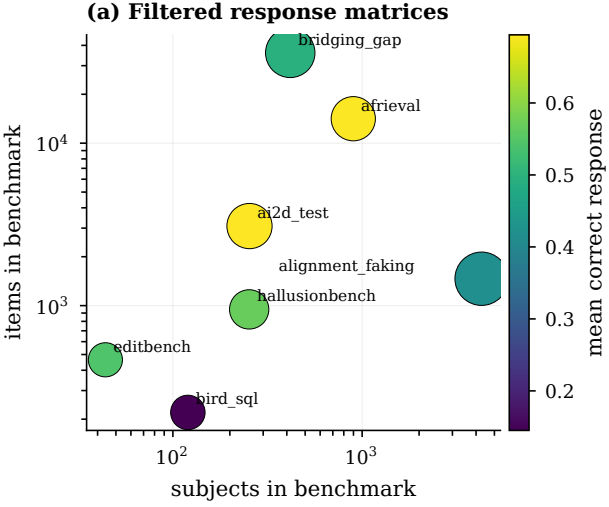




The narrow predicted distributions and near-zero within-benchmark correlations motivate the filtered, benchmark-centred holdout work. These configurations did not transfer item difficulty reliably.

What is analysed in this appendix

Two different objects are kept separate: observed binary responses and frozen item-text embeddings.



Response graph

A bipartite edge joins a subject to an item only when that binary response was observed. Missing responses are not imputed.

Qwen item graph

Each filtered item connects to its 20 nearest 1024-d Qwen vectors by cosine similarity. It uses question text only.

Why separate them

Response structure needs labels and is training-only. Qwen edges can be built for a new item before its label is known.

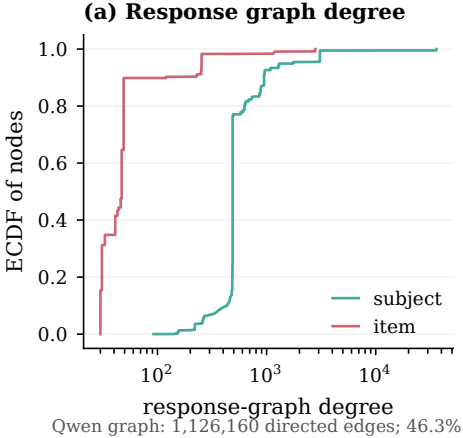
Filtering: 16,278,271 raw rows to 5,150,869; item support >= 30, subject-task responses >= 80.

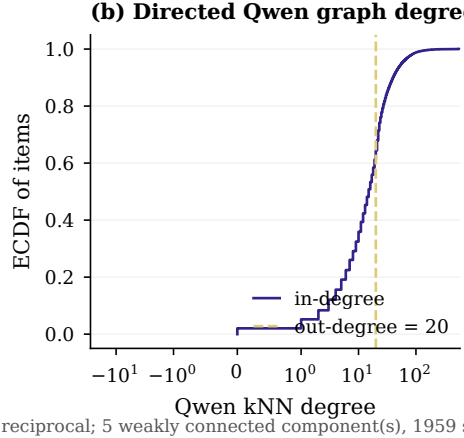
How the two graphs are constructed

The response graph and the Qwen kNN graph have different edges, degree distributions, and test-time sta



Direction comes from the source item's own Top-K list. It need not be mutual: $i \to j$ can exist while $j \not\to i$ doe

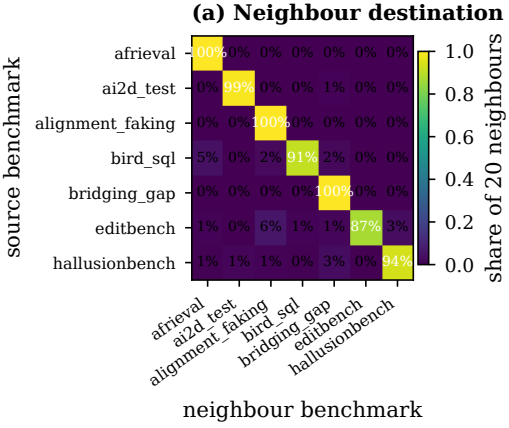


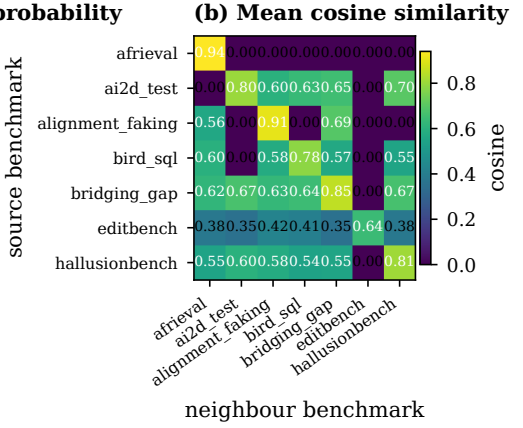


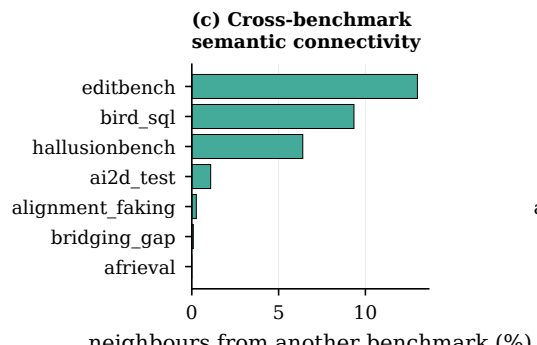
Qwen graph: 1,126,160 directed edges; 46.3% are reciprocal; 5 weakly connected component(s), 1959 strongly

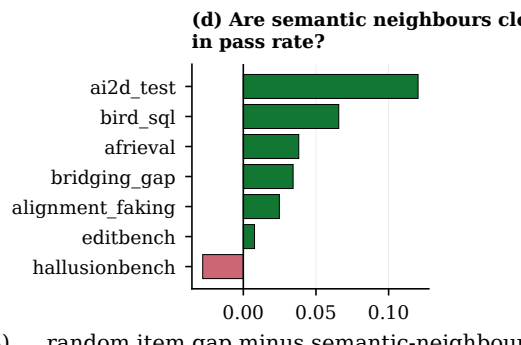
Qwen3 item similarity forms a content-only graph

Each row contains the 20 nearest text-embedding neighbours of items from that benchmark. It is directed
each item chooses its own neighbours.





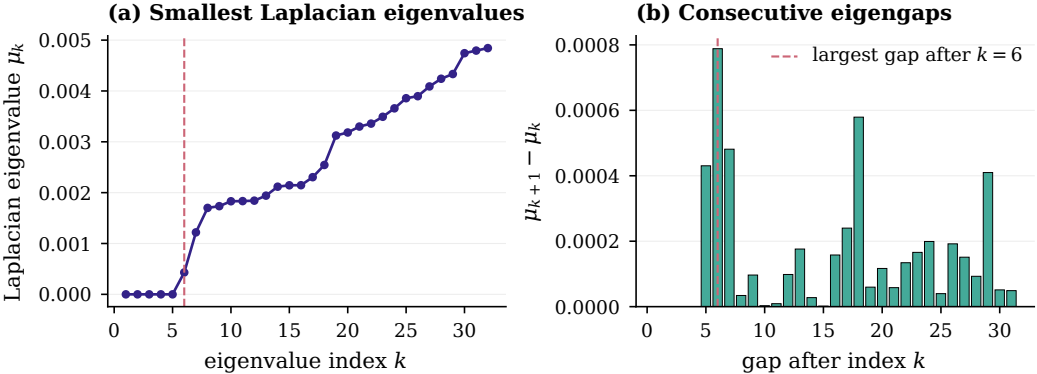




Panel (d): positive means Qwen neighbours have more similar observed pass rates than random items, after matching so

Laplacian spectrum of the symmetrised Qwen kNN graph

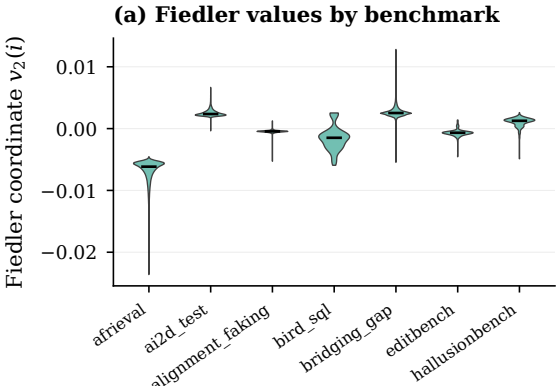
The graph symmetrises directed Top-20 cosine edges. Near-zero values count components; an eigengap is clustering heuristic.

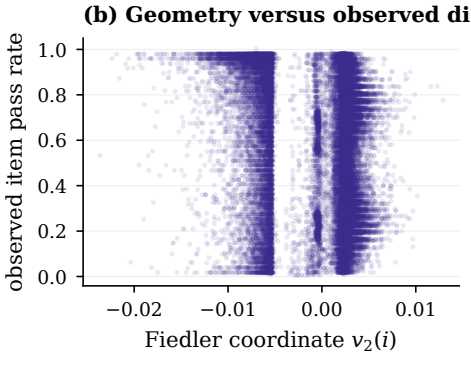


There are 5 weak components, so the first 5 eigenvalues should be near zero. The largest displayed gap is after k=6.

What the Fiedler vector can, and cannot, show

It is the second-smallest-eigenvalue vector, computed inside the largest weak component; the disconnecte
graph has no unique Fiedler vector.





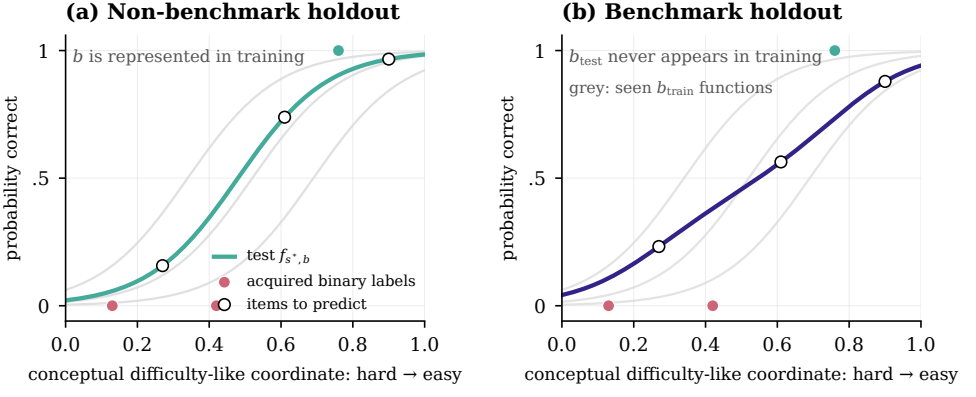
Largest weak component: 56,221 items. The sign of v_2 is arbitrary. Separation by benchmark means the graph retain

What is a function in the two evaluation protocols?

One function is one subject--benchmark task. Curves are conceptual, not fitted IRT estimates.

Task $\tau = (s, b)$: $D_{s,b} = \{(e_i, y_{s,b,i})\}_i$ is a finite sample from $f_{s,b}(e_i) = \Pr(Y_{s,b,i} = 1 \mid e_i, s, b)$.

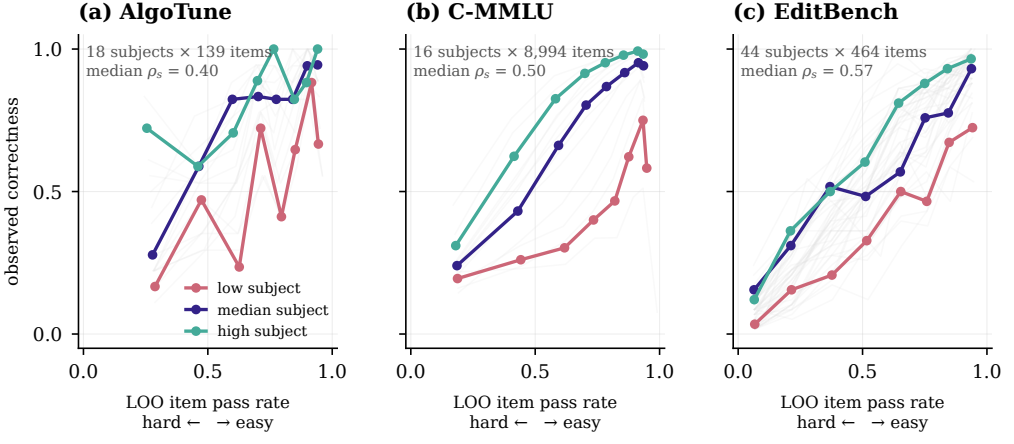
The NP receives a few labelled context items from this task and predicts held-out items. At test time it changes its conditional prediction through the context, not its network weights.



Grey curves sketch functions from a familiar training benchmark. In (a), the test task shares that benchmark's item family and response structure. In (b), the test function belongs to an unseen benchmark, so the conditional prior must transfer through item content embeddings. The horizontal coordinate is only an explanatory slice: the evaluated NP uses high-dimensional item embeddings, not a known scalar difficulty.

Observed subject response functions on held-out benchmarks

Post-hoc only: the x-axis uses labels from other test subjects.



Faint traces are all held-out test tasks. Coloured curves select the subjects nearest the 10th, 50th, and 90th percentiles of observed task accuracy. Each point averages that subject's binary responses in one of eight equal-count bins of leave-one-subject-out item pass rate. These reveal real response-function variation, but use held-out benchmark labels and are strictly post-hoc.